CU-TBI FORMAL INCUBATION & PRE-SEED GRANT PROPOSAL

Project: Autonomous GPS-Denied Vision-Based UAV Localization System (NavZero / Project Trishul)

Applicant: Student Innovator, Chandigarh University | **Target:** CU-TBI Pre-Incubation / NIDHI-PRAYAS / NIDHI-EIR
Domain: Aerospace & Defense / Autonomous Robotics | **Readiness:** TRL 3/4 (Analytical Prototype Complete)

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1. EXECUTIVE SUMMARY

Electronic Warfare (EW) and localized Global Navigation Satellite System (GNSS) jamming have emerged as the primary failure point for tactical Unmanned Aerial Vehicles (UAVs) in modern contested border sectors (e.g., Eastern Ladakh). Commercial-off-the-shelf and standard military UAVs relying on satellite positioning experience catastrophic navigational loss, drifting, or mission abortion when GPS signals are suppressed.

Project Trishul / NavZero is an indigenous, edge-compute visual localization software engine that enables tactical UAVs to autonomously determine their global geodetic coordinates (X, Y) and execute autonomous Return-to-Home (V-RTH) maneuvers with **zero dependence on GPS, magnetometer compasses, or external radio-frequency emissions**.

Current Technical Accomplishments (Validated on Consumer Quad-Core CPU):

- **Real-time execution:** 26–45 frames per second on low-power quad-core CPU architecture without requiring bulky dedicated GPUs.
- **Algorithmic architecture:** Multi-scale ORB binary descriptors, bitwise Hamming matching, Lowe's distance ratio filtering (d1 < 0.75 * d2), and projective RANSAC geometric outlier rejection.
- **Geospatial integration:** Standard 6-parameter GDAL/USGS affine coordinate transformations mapping camera frame pixels to metric map coordinates.
- **Software rigor:** 34 passing automated unit tests (100% pass rate) with a sub-250MB memory footprint.

*We are seeking **formal pre-incubation, laboratory workspace access, and prototype grant endorsement (NIDHI-PRAYAS / NIDHI-EIR)** through CU-TBI to transition this proven software engine onto an embedded physical airframe (Raspberry Pi 5 + Pixhawk 6C) for outdoor flight validation.*

2. PROBLEM STATEMENT & NATIONAL DEFENSE RELEVANCE

In contested environments, adversary ground vehicles deploy high-power Electronic Warfare (EW) jammers broadcasting broadband noise across L1/L2 satellite bands. Because satellite signals originate 20,000 km in space, their ground received power is exceptionally weak (~ -160 dBW). A low-cost 50-watt ground jammer effectively denies GPS across a 20–30 km radius.

When GPS fails, conventional drones enter fail-safe descent inside hostile territory or drift uncontrolled into mountain cliffs. Inertial Measurement Units (IMUs) accumulate quadratic drift over time (~ 1/2 * a * t^2), deviating by hundreds of meters in minutes. The Indian Ministry of Defence (MoD) and Innovations for Defence Excellence (iDEX) have identified **GPS-Denied UAV Navigation** as a high-priority challenge under recent Defence India Startup Challenges (DISC). Our project directly addresses this critical capability gap under *Aatmanirbhar Bharat*.

3. THE PROPOSED SOLUTION: NAVZERO OPTICAL ARCHITECTURE

Our system provides a passive, non-emissive optical autopilot that navigates by matching physical terrain passing beneath the drone's belly camera against pre-loaded reconnaissance orthophoto tiles stored in onboard memory, operating in two distinct modes:

- **Mode A (Absolute Map Matching):** When satellite/aerial imagery is pre-loaded before takeoff, the drone computes exact geodetic coordinates (Lat, Lon) by locking onto permanent geological landmarks at 30 Hz update rates.
- **Mode B (Zero-Map Visual Breadcrumb Return):** When flying into uncharted territory with zero maps, the engine tracks inter-frame

visual breadcrumbs along its outbound flight path. When GPS is jammed or radio contact severed, the autopilot engages Autonomous Visual Return-to-Home (V-RTH), reversing heading and retracing its visual footprints all the way back to the exact takeoff point.

4. BUDGET & BILL OF MATERIALS (BOM) FOR CU-TBI / NIDHI-PRAYAS

To transition this technology from **TRL 4 (Laboratory Validation)** to **TRL 6 (Physical Airframe Demonstration)**, we request temporary issue of existing college lab components or prototype grant funding under NIDHI-PRAYAS according to the following itemized Bill of Materials:

Item Description	Qty	Technical Purpose	Est. Cost (■)
Raspberry Pi 5 (8GB) + Active Cooler	2	Onboard companion computer running Python/OpenCV vision pipeline	■18,000
Pixhawk 6C / 4 Flight Controller Kit	1	Open-source autopilot running ArduPilot for MAVLink telemetry fusion	■16,000
Global Shutter Downward Camera Module	2	High-speed, distortion-free optical ground tracking sensor	■8,000
F450 Quadcopter Airframe + Motors/ESCs	1	Basic quadcopter test bench for outdoor flight integration	■14,000
LiPo Batteries (4S 5000mAh) + Smart Charger	2	Propulsion and avionics power supply	■12,000
Benchtop Rig Hardware & Connectors	-	Acrylic mounting plates, FTDI cables, telemetry radios	■7,000
Field Testing & Safety Compliance	-	Safety nets, ground calibration markers, field logistics	■10,000
TOTAL ESTIMATED PROTOTYPE BUDGET:		**Complete 6-Month Physical Airframe Integration**	**■85,000**

Note: If CU robotics/aerospace laboratories already possess Pixhawk controllers and Raspberry Pis, the required cash grant can be reduced to under ■25,000 for consumables.

5. 6-MONTH MILESTONE EXECUTION ROADMAP

Phase	Timeline	Key Technical Deliverables & Milestones
Phase 1: Embedded Porting	Month 1–2	Port existing Python/OpenCV vision codebase to Raspberry Pi OS (Linux ARM64). Benchmark C++ bindings to achieve stable 30+ FPS at under 7 watts power consumption.
Phase 2: Hardware-in-Loop	Month 3–4	Establish two-way UART serial communication between Raspberry Pi and Pixhawk flight controller using pymavlink. Verify simulated GPS failsafe handoff on benchtop rig without propellers.
Phase 3: Tethered Flight	Month 5	Integrate hardware onto F450 quadcopter airframe. Conduct low-altitude tethered hover tests over college ground markers. Demonstrate GPS jamming injection and automatic optical position holding.
Phase 4: Full Outdoor RTH & iDEX Filing	Month 6	Execute full outdoor autonomous mission: manual takeoff -> GPS cut -> autonomous optical return-to-home and landing. Film high-resolution video proof and submit formal application for iDEX DISC (■1.5 Cr) and Smart India Hackathon under Chandigarh University's banner.

6. COMMERCIAL OPPORTUNITY & PRESTIGE FOR CHANDIGARH UNIVERSITY

1. **Defense Procurement:** The primary customer is the Indian defense ecosystem (Indian Army artillery units, Northern Command surveillance, BSF border patrol). By developing an export-free, indigenous software kit, we can retrofit existing military drone fleets with GPS-denied survivability.
2. **Institutional Prestige for CU:** Submitting a validated TRL-6 defense project to national platforms (iDEX, DefTech, SIH) significantly enhances Chandigarh University's NIRF research ranking, NAAC accreditation, and national media recognition.
3. **Immediate Live Demo:** An interactive physical demonstration of the computer vision engine, flight simulator, and 34-test suite can be delivered to the CU-TBI screening committee on 10 minutes notice.

Submitted By:	**Endorsed & Forwarded By:**	**Received By CU-TBI:**
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Student Innovator / Founder Chandigarh University, Punjab	**Head of Department / Faculty Mentor** Chandigarh University	**Incubation Manager / In-Charge** Technology Business Incubator (CU-TBI)